

LEEK THORNCLIFFE, United Kingdom

WMO: 033300

Lat: 53.1278N

Lon: 1.9814W

Elev: 299

StdP: 97.78

Time Zone: 0.00 (EUW)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	-2.8	-6.2	2.3	-2.2	-5.0	2.6	-1.6	13.9	5.4	12.8	4.5	4.1	30	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	23.7	17.8	21.6	16.7	20.0	16.0	18.5	22.1	17.5	20.5	16.5	19.1	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.7	19.9	16.2	12.0	18.8	15.5	11.4	17.7	53.6	22.2	50.2	20.6	47.4	19.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.5	11.2	10.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-5.7	26.7	1.7	2.0	-6.9	28.1	-7.9	29.3	-8.9	30.4	-10.1	31.9
			-6.2	20.6	1.6	1.2	-7.4	21.4	-8.3	22.1	-9.2	22.8	-10.4	23.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	3.2	3.3	4.9	7.5	10.3	13.2	15.2	14.4	12.5	9.6	5.7	3.9	
	DBStd	5.13	2.84	3.09	3.06	2.97	3.04	2.67	2.73	2.12	2.44	2.67	2.97	3.23	
	HDD10.0	1041	212	188	159	86	33	3	0	0	5	38	130	188	
	HDD18.3	3536	471	421	416	325	250	154	106	124	175	270	378	447	
	CDD10.0	559	0	1	2	11	41	100	160	136	81	26	2	1	
	CDD18.3	11	0	0	0	0	0	2	8	1	1	0	0	0	
	CDH23.3	67	0	0	0	0	1	13	46	6	1	0	0	0	
	CDH26.7	4	0	0	0	0	0	0	4	0	0	0	0	0	
Wind	WSAvg	4.7	5.3	5.1	5.2	5.1	4.8	4.3	4.0	3.9	4.3	4.9	4.8	4.9	
Precipitation	PrecAvg	1031	96	81	66	71	71	80	80	83	80	109	103	112	
	PrecMax	1375	195	193	144	164	131	197	190	241	158	239	221	198	
	PrecMin	777	8	18	10	10	15	20	34	21	16	36	45	28	
	PrecStd	157	46	39	34	38	32	44	34	42	36	48	40	42	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.8	14.7	19.3	22.5	25.2	26.7	24.1	21.8	17.9	13.8	11.9	
		MCWB	10.5	9.4	10.6	13.1	16.4	17.9	18.9	18.1	17.7	15.2	13.2	11.0	
	2%	DB	9.8	9.8	12.2	15.8	19.9	22.1	24.5	20.9	19.3	15.6	12.3	10.7	
		MCWB	9.2	8.2	9.3	11.2	15.4	16.7	18.2	16.7	16.0	14.3	11.7	10.2	
	5%	DB	8.5	8.7	10.6	13.8	17.5	20.2	22.5	19.4	17.7	14.4	11.2	9.7	
		MCWB	8.0	7.7	8.8	10.2	13.5	15.8	17.3	16.1	15.0	13.4	10.6	9.2	
	10%	DB	7.5	7.6	9.2	12.2	15.3	18.3	20.4	18.1	16.4	13.4	10.0	8.6	
		MCWB	7.0	6.8	7.8	9.4	12.0	14.5	16.3	15.2	14.2	12.4	9.5	8.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	9.7	11.1	14.0	17.3	19.6	20.2	19.0	18.2	15.9	13.2	11.2
			MCDB	10.9	10.7	13.1	18.2	21.1	23.3	24.9	22.2	20.6	17.2	13.8	11.5
2%		WB	9.3	8.8	10.0	12.0	15.5	17.4	18.9	17.7	16.8	14.6	11.8	10.3	
		MCDB	9.7	9.7	11.5	14.9	19.4	20.7	23.2	20.0	18.7	15.5	12.2	10.7	
5%		WB	8.1	7.9	9.1	10.9	14.0	16.1	17.7	16.7	15.6	13.7	10.7	9.3	
		MCDB	8.4	8.5	10.4	13.2	16.8	19.4	21.3	18.6	17.0	14.3	11.1	9.7	
10%		WB	7.0	6.9	8.0	9.8	12.7	15.0	16.7	15.9	14.7	12.6	9.6	8.1	
		MCDB	7.4	7.5	9.1	12.1	14.8	17.6	19.7	17.4	15.9	13.2	9.9	8.6	
Mean Daily Temperature Range		5% DB	MDBR	4.1	4.4	5.8	7.0	7.5	7.6	7.4	6.6	6.3	5.1	4.5	4.0
			MCDBR	5.4	5.3	7.0	8.9	10.0	10.3	10.8	8.2	7.7	5.5	5.3	5.1
	MCWBR		5.6	4.5	4.8	5.7	6.5	6.2	5.9	4.8	4.9	4.3	5.1	4.9	
	5% WB	MCDBR	5.4	5.0	6.5	8.1	9.1	9.4	9.8	7.2	7.0	5.3	5.4	5.1	
		MCWBR	5.6	4.5	5.0	5.8	6.3	6.2	5.9	4.8	5.1	4.6	5.2	5.0	
Clear-Sky Solar Irradiance	taub	0.313	0.328	0.357	0.382	0.385	0.384	0.384	0.370	0.355	0.337	0.314	0.301		
	taud	2.351	2.401	2.321	2.263	2.292	2.316	2.317	2.376	2.424	2.450	2.410	2.323		
	Ebn at Noon	663	779	824	847	864	866	859	851	821	757	661	599		
	Edh at Noon	59	76	100	120	122	121	119	106	90	71	55	50		
All-Sky Solar Radiation	RadAvg	0.68	1.36	2.47	3.81	4.78	5.01	4.79	3.94	2.90	1.64	0.83	0.54		
	RadStd	0.06	0.15	0.27	0.41	0.32	0.48	0.50	0.27	0.24	0.14	0.09	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page